

Generation of Functional Hepatocytes from Human Embryonic Stem Cells Under Chemically Defined Conditions that Recapitulate Liver Development

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Generation of hepatocytes from human embryonic stem cells (hESCs) could represent an advantageous source of cells for cell therapy approaches as an alternative to orthotopic liver transplantation. However, the generation of differentiated hepatocytes from hESCs remains a major challenge, especially using a method compatible with clinical applications. We report a novel approach to differentiate hESCs into functional hepatic cells using fully defined culture conditions, which recapitulate essential stages of liver development. hESCs were first differentiated into a homogenous population of endoderm cells using a combination of activin, fibroblast growth factor 2, and bone morphogenetic protein 4 together with phosphoinositide 3-kinase inhibition. The endoderm cells were then induced to differentiate further into hepatic progenitors using fibroblast growth factor 10, retinoic acid, and an inhibitor of activin/nodal receptor. After further maturation, these cells expressed markers of mature hepatocytes, including asialoglycoprotein receptor, tyrosine aminotransferase, α 1-antitrypsin, Cyp7A1, and hepatic transcription factors such as hepatocyte nuclear factors 4 α and 6. Furthermore, the cells generated under these conditions exhibited hepatic functions *in vitro*, including glycogen storage, cytochrome activity, and low-density lipoprotein uptake. After transduction with a green fluorescent protein-expressing lentivector and transplantation into immunodeficient uPA transgenic mice, differentiated cells engrafted into the liver, grew, and expressed human albumin and α 1-antitrypsin as well as green fluorescent protein for at least 8 weeks. In addition, we showed that hepatic cells could be generated from human-induced pluripotent cells derived from reprogrammed fibroblasts, demonstrating the efficacy of this approach with pluripotent stem cells of diverse origins. **Conclusion:** We have developed a robust and efficient method to differentiate pluripotent stem cells into hepatic cells, which exhibit characteristics of human hepatocytes. Our approach should facilitate the development of clinical grade hepatocytes for transplantation and for research on drug discovery. (HEPATOLOGY 2010;51:1754-1765.)

Abbreviations: AAT, α 1-antitrypsin; AFP, alpha-fetoprotein; APOA-II, apolipoprotein A-II; BMP, bone morphogenetic protein; CDM, chemically defined medium; CK, cytokeratin; CT, cycle threshold; DEC, definitive endoderm cell; EGF, epidermal growth factor; FBS, fetal bovine serum; GFP, green fluorescent protein; FACS, fluorescent-activated cell sorting; FGF, fibroblast growth factor; hESC, human embryonic stem cell; HGF, hepatocyte growth factor; hiPSC, human-induced pluripotent stem cell; HNF, hepatocyte nuclear factor; ICG, indocyanine green; PBGD, porphobilinogen deaminase; PI3K, phosphoinositide 3-kinase; PVA, polyvinyl alcohol; Q-PCR, quantitative polymerase chain reaction; RA, retinoic acid; RT-PCR, reverse-transcription polymerase chain reaction.

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Liver diseases are becoming one of the most common causes of mortality in developing countries, and orthotopic liver transplantation is the only available treatment. However, an increasing number of patients die while on the liver transplant waiting list due to the shortage of suitable donor livers.¹ Hepatocyte transplantation recently became an alternative to orthotopic liver transplantation for the treatment of acute failure and life-threatening metabolic liver diseases.² However, this strategy is also restricted by the lack of donors and by the limited number of cells, because functional human hepatocytes cannot be expanded *in vitro* and are difficult to cryopreserve.³ Therefore, exploring other sources of cells, either of hepatic or extrahepatic origin, to generate hepatic cells capable to proliferate *in vitro* and to express hepatic-specific functions remains a major goal. Recently, human hepatic stem cells isolated from adult livers were shown to be functional *in vivo* after transplantation into a mouse model of liver injury.⁴ More recently, human adipose tissue-derived mesenchymal stem cells were also differentiated into hepatocyte-like cells *in vitro* and were able to engraft and express hepatic-specific proteins in recipient mouse livers.⁵ However, the number of cells that need to be expanded *in vitro* and differentiated still constitutes a limitation to these approaches.

Human pluripotent stem cells represent an alternative source of hepatocytes to solve the lack of liver cells for regenerative medicine. They can be derived from embryos at the blastocyst stage as human embryonic stem cells (hESCs)⁶ or from fibroblasts reprogrammed by overexpressing transcription factors such as Oct-4, Sox2, KLF-4, and C-Myc (human-induced pluripotent stem cells [hiPSCs]).^{7,8} hESCs proliferate infinitely *in vitro* while maintaining the capacity to differentiate into a broad number of cell types, and recent results suggest that human induced pluripotent stem cells (hiPSC) share these characteristics.⁹ Several groups have already reported the differentiation of hESCs into hepatic cells using diverse culture systems.¹⁰⁻¹⁵ However, these approaches are all based on culture media containing serum, complex matrices such as matrigel, and/or mouse embryonic fibroblasts as feeders—all of which are sources of unknown

factors that could obscure studies of molecular mechanisms controlling human liver development or render the resulting tissues incompatible with future clinical applications. In this study, we describe a new culture system for driving differentiation of hESCs into hepatocytes using fully defined culture media devoid of animal products or unknown factors. This approach follows a natural path of development by respecting key stages of liver development, which may provide the best approach for generating differentiated cells with native properties. The differentiated hepatic cells expressed specific adult liver cell markers and displayed hepatic functions *in vitro*. We also showed that these cells were capable to participate in liver growth *in vivo* after transplantation into newborn uPAxrag2gammac^{-/-} mouse liver. Finally, our approach also induced differentiation of hiPSCs generated from human fibroblasts.

Materials and Methods

Cell Culture. hESCs (H9; WiCell, Madison, WI) were grown in chemically defined medium (CDM) as described.¹⁶

Fetal livers were obtained after termination of pregnancy at 11-13 weeks' gestation, with the informed consent of the mothers, as recommended by the French Ethics Committee and the local Ethics Committee of Paris XI University (Paris, France). Human hepatoblasts were isolated and cultured as described.¹⁷ Primary human hepatocytes were isolated from normal liver tissue biopsies obtained during resections (partial hepatectomies) for hepatic or metastatic tumors after informed patient consent. The primary hepatocytes were obtained by a two-step collagenase method and cultured as described.¹⁷

Differentiation of hESCs into Hepatic Cells. Cells were harvested using 5 mg/mL collagenase IV (Invitrogen) and then plated into plates (Costar; Corning Life Sciences, Acton, MA) precoated either with fetal bovine serum (FBS) or with 15 g/mL of human Fibronectin (Chemicon, Molsheim, France). Fibronectin-coated plates were used for small-scale experiments (see Figs. 1, 2, and 4), and FBS-coated plates were used in large-scale experiments (see Figs. 5, 6, and 7). No differences were

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Additional Supporting Information may be found in the online version of this article.

observed between cells differentiated on fibronectin-coated plates and those differentiated on FBS-coated plates.

The following three days cells were grown in presence of Ly294002 (Calbiochem), activin A, fibroblast growth factor (FGF) 2, and bone morphogenetic protein (BMP) 4 (R&D Systems). To induce hepatic endoderm, cells were then cultured during 3 days in the presence of FGF10 (Autogenbioclear, Nottingham, UK) and then in the presence of retinoic acid (RA) (Sigma), SB431542 (Tocris, Bristol, UK), and FGF10 (Autogenbioclear) for 2 days. Finally, the cell population was grown in CDM supplemented with FGF4, hepatocyte growth factor (HGF), and epidermal growth factor (EGF) (Peprotech, Neuilly-Sur-Seine, France) for 10 days.

Reverse-Transcription and Quantitative Polymerase Chain Reaction. Reverse-transcription polymerase chain reaction (RT-PCR) and quantitative PCR (Q-PCR) experiments were performed as described.¹⁸

The results are presented as the mean of three independent experiments and error bars indicate standard deviation. The number of Q-PCR cycles (cycle threshold [CT] value) is also indicated for key genes and for porphobilinogen deaminase (PBGD).

Microarray Analyses. Specific experimental details are provided in the Supporting Information.

Immunofluorescence, Flow Cytometry, and Immunohistochemistry. These assays relied on well-established procedures. Details are provided in the Supporting Information.

Lentivirus Production and Transduction of hESCs. The EF1 α -green fluorescent protein (GFP) lentivector was constructed and produced by Vectalys (Toulouse, France). The apolipoprotein A-II (APOA-II)-GFP lentivector was constructed in the laboratory and produced by Vectalys.

Prior to transduction with lentiviruses, hESCs were dissociated and were incubated with viral particles for 3 hours at 37°C in low-attachment 24-well plates (Corning Life Sciences) under gentle rocking before being seeded onto mitotically inactivated mouse embryonic fibroblasts in hESC medium containing 4 ng/mL FGF2 (R&D Systems). Undifferentiated transduced cells were expanded and differentiated using chemically defined conditions described above.

Animals. Animal studies were conducted under protocols approved by the French Ministry of Agriculture. Differentiated cells (5×10^5 cells/animal in 50 μ L saline solution) were injected into the liver of 5-day-old uPAxRag2gammac^{-/-} mice (n = 9). To control non-adaptive defense, mice were treated as described.¹⁹ The transplanted mice were killed 8 weeks after transplanta-

tion. Blood samples were collected, and human albumin was quantified in sera by enzyme-linked immunosorbent assay. Livers were removed for histologic analysis, and liver fragments were either embedded in OCT compounds and then frozen in liquid nitrogen, or were fixed in 4% paraformaldehyde and embedded in paraffin. Non-transplanted mice were used as controls.

Results

Combination of Activin, FGF2, BMP4, and LY294004 Drives Differentiation of hESCs into Definitive Endoderm. Our differentiation strategy is illustrated in Fig. 1.

During organogenesis in mammalian embryos, the liver is specified from the ventral foregut endoderm and consequently, endoderm cells represent the earliest progenitors of hepatoblasts. To induce differentiation of hESCs into definitive endoderm cells (DECs), we analyzed the effects of three factors known to control specification of the three germ layers in mammals: activin A, BMP4, and FGF2.²⁰ This analysis was performed in the absence of feeder cells using CDM,²¹ which is devoid of serum and is supplemented with insulin, transferrin, and defined lipids to which polyvinyl alcohol (PVA) was added as a substitute for bovine serum albumin. H9 and hSF-6 hESC lines were plated on plastic dishes precoated with human fibronectin or serum²² and grown in CDM supplemented with activin A (10 ng/mL) and FGF2 (12 ng/mL) for 2 days. Cells were then cultured in the presence of diverse combinations of the three factors for 5 days, and expression of the pluripotency marker (Oct-4), mesendoderm marker (Brachyury), and definitive endoderm marker (Sox17) were analyzed using Q-PCR (Supporting Fig. 1). The assessment of the three growth factors showed that a high dose of activin A (100 ng/mL) was necessary and sufficient to induce the expression of endoderm markers Sox17, GSC, and Mixl1 in hESCs (Fig. 2A), but pluripotency markers such as Oct-4 and Nanog remained expressed (Fig. 2A, Supporting Fig. 2). These results are in agreement with recent publications showing that activin/nodal signaling controls the expression of key pluripotent transcription factors such as Nanog in hESCs.^{23,24} They also suggest that, in our conditions, activin A is sufficient to induce the expression of endoderm markers in hESCs and that cofactors are required to block its effect on pluripotency.

Combination of high doses of activin A, FGF2, and BMP4 induced the strongest increase in endoderm markers, but none of the combinations analyzed was efficient to decrease the expression of pluripotency markers (Fig. 2B). We then investigated the effect of phosphoinositide

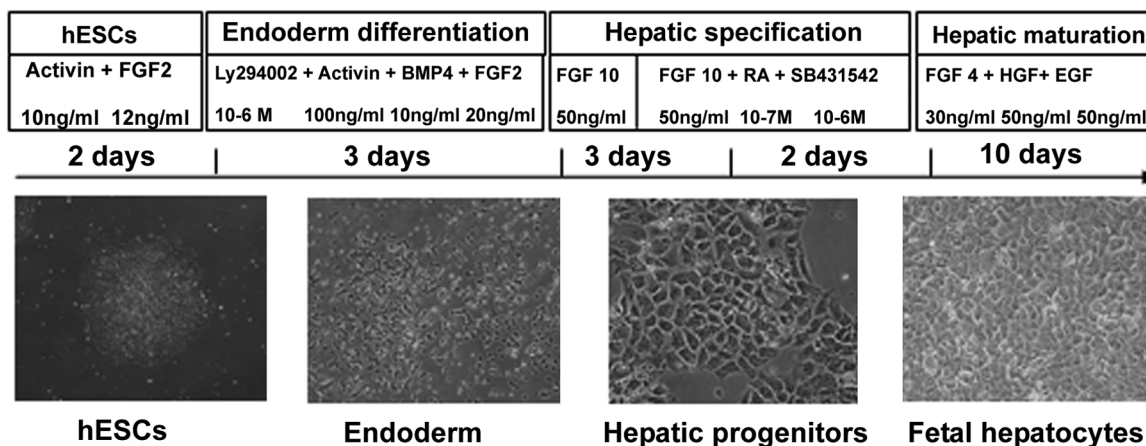


Fig. 1. Protocol to differentiate pluripotent human cells into hepatocytes. H9 cells or hiPSCs were passaged and grown for 2 days in CDM + 10 ng/mL activin + FGF2 12 ng/mL. hESCs were then grown for 3 days in culture conditions inductive for endoderm differentiation (CDM/PVA, 100 ng/mL activin, 20 ng/mL FGF2, 10 ng/mL BMP4, 10 μ M LY294003). Endoderm cells were further differentiated for 3 days into anterior definitive endoderm using 50 ng/mL FGF10. Hepatic progenitors were obtained by growing the resulting anterior definitive endoderm cells for 2 days in CDM/PVA, 50 ng/mL FGF10, 10⁻⁷ M retinoic acid, 10 μ M SB431542. Hepatocytes were obtained after 10 additional days in CDM/PVA, 30 ng/mL FGF4, 50 ng/mL HGF, and 50 ng/mL EGF.

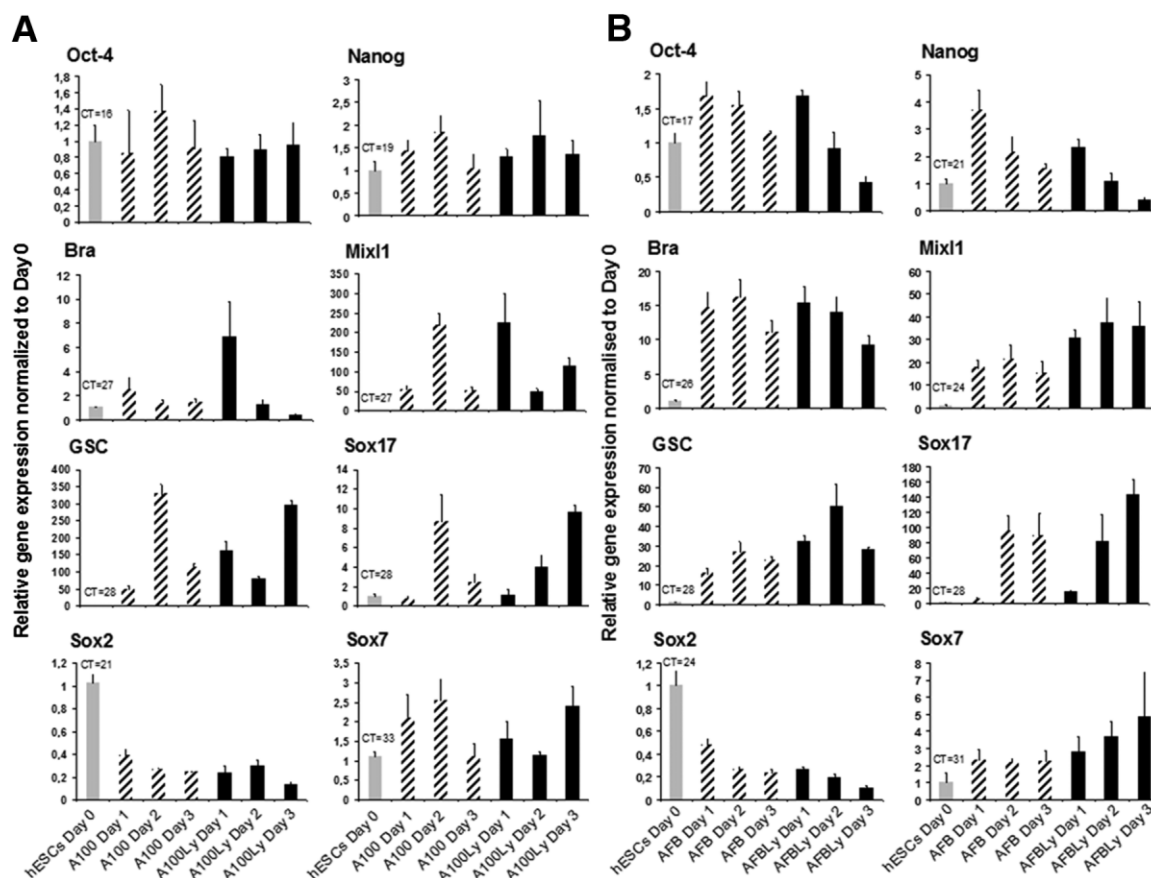


Fig. 2. Combination of high doses of activin, FGF2, BMP4, and inhibition of PI3K drives hESCs to differentiate into a homogenous population of definitive endoderm cells. (A) Inhibition of PI3K in the presence of high doses of activin is not sufficient to induce differentiation of hESCs into endoderm. H9 cells were grown for 3 days in the presence of activin (100 ng/mL) $\pm$ the PI3K inhibitor LY29002 (10 μ M). RNAs were extracted every day, and expression of the denoted genes was determined by way of Q-PCR. CT of PBGD, 24 $\pm$ 1. (B) H9 cells were grown for 3 days in the presence of activin (100 ng/mL), FGF2 (20 ng/mL), and BMP4 (10 ng/mL) $\pm$ LY29002 (10 μ M). Expression of the denoted genes was determined by way of Q-PCR. CT of PBGD, 24 $\pm$ 1.

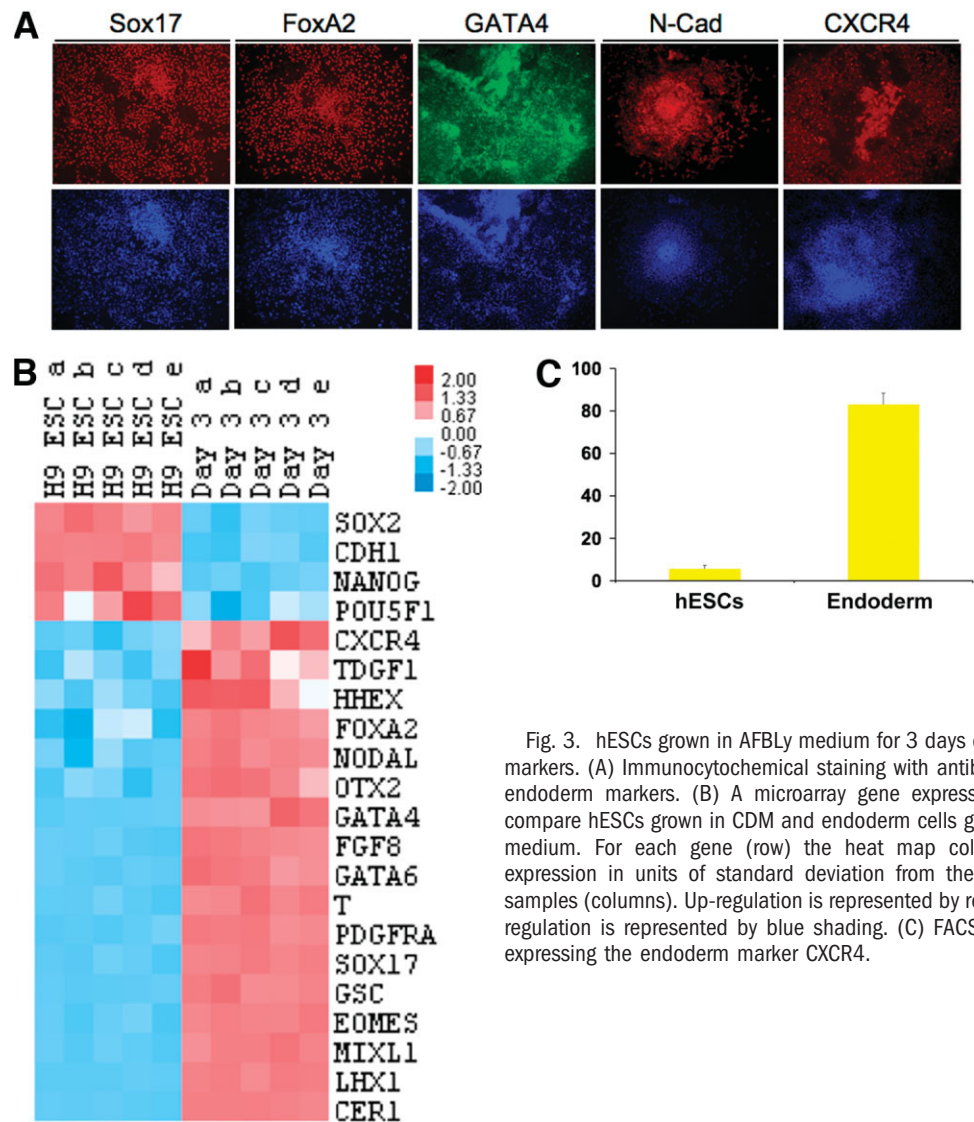


Fig. 3. hESCs grown in AFBly medium for 3 days express endoderm markers. (A) Immunocytochemical staining with antibodies specific for endoderm markers. (B) A microarray gene expression heat map to compare hESCs grown in CDM and endoderm cells generated in AFBly medium. For each gene (row) the heat map colors sample gene expression in units of standard deviation from the mean across all samples (columns). Up-regulation is represented by red shading; down-regulation is represented by blue shading. (C) FACS analysis of cells expressing the endoderm marker CXCR4.

3-kinase (PI3K) inhibition on hESCs, because it has been shown to increase the effect of activin/nodal signaling on endoderm differentiation of hESCs grown on feeders and on medium containing serum.²⁵ Combination of high doses of activin A and LY294002, a chemical inhibitor of PI3K, resulted in a significant increase in expression of endoderm markers (Fig. 2A), confirming that the PI3K pathway might have an inhibitory effect on endoderm differentiation. However, addition of LY294002 was not sufficient to repress the expression of pluripotency markers (Fig. 2A), suggesting that this inhibitor did not interfere with the mechanisms involving activin/nodal signaling in pluripotency. Full differentiation was only achieved by combining high doses of activin A, BMP4, FGF2, and LY294004 in CDM (AFBly medium) (Fig. 2B), suggesting that only the combination of different signals of differentiation can block the positive effect of activin/nodal signaling on pluripotency.

The cells generated in AFBly medium expressed definitive endoderm markers (Sox17, GSC, Mixl1, Lhx1, CXCR4, GATA6, Eomes, Hex, and so forth) as assessed by Q-PCR, immunostaining, and gene expression profiling (Figs. 2B, 3A-C). Importantly, the expression of extra-embryonic markers such as Sox7 (Fig. 2B) and the expression of neuroectoderm markers such as Sox2 (Figs. 2B, 3B) were not detected, excluding that cell types generated under these culture conditions could be primitive endoderm cells or neuronal progenitors. Finally, fluorescence-activated cell sorting (FACS) analysis revealed that 80% of the cells generated in AFBly medium expressed the endoderm marker CXCR4, suggesting that our approach allowed the differentiation of hESCs into near homogenous population of endoderm cells (Fig. 3C).

Differentiation of Human Endoderm Cells into Hepatic Progenitors. We then investigated the capacity

of DEC cells generated in AFB₁ medium to differentiate further into liver progenitors in the presence of diverse combinations of factors known to be involved in the patterning of the primitive gut, including FGF10, Wnt, RA, and activin A. The effect of inhibiting the activin and Wnt pathways was also analyzed using SB431542, a pharmacological inhibitor of activin/nodal receptors,²⁶ or Frizzled-8/Fc chimera, an inhibitor of the Wnt signaling pathway.²⁷

Generation of liver cells was monitored for the expression of hepatocyte nuclear factor (HNF) 4 α and α -fetoprotein (AFP) (Fig. 4A,B), two markers expressed in hepatic progenitors during the early stages of liver development. The highest induction in HNF4 α and AFP expression was observed when DEC cells were first grown for 3 days in the presence of FGF10, and then for 2 additional days in a combination of FGF10, RA, and SB431542 (Figs. 1, 4A-C). Inhibition of Wnt signaling decreased hepatic differentiation, which confirms recent observations showing that Wnt has an essential function in the mechanisms controlling the differentiation of DEC cells into hepatic endoderm.¹⁵ However, exogenous Wnt was not necessary in our culture conditions, because DEC cells already expressed a high level of Wnt3a (Supporting Fig. 3). These results suggest that RA plays an important role in hepatic specification, which is synergized by FGF10 and Wnt signaling, whereas transforming growth factor β signaling might interfere with the early step of DEC differentiation into liver progenitors.

Most of the hepatic progenitors generated by the combination of FGF10, RA and SB431542 expressed EpCAM (Fig. 4D). HNF4 α and AFP or cytokeratin (CK) 19 were coexpressed in 60% and 50% of the cells, respectively (Fig. 4C,D). Because hepatocytes and biliary epithelial cells derive from a common bipotential progenitor (hepatoblasts) during liver development, our results suggest that a population of hepatoblasts has been generated during the differentiation process.

Maturation of Hepatic Progenitors into Hepatocyte-Like Cells. We then developed culture conditions to differentiate hepatic progenitors into hepatocytes using factors known to be involved in these mechanisms *in vivo*. We observed that the combination of FGF4, HGF, and EGF was sufficient to drive differentiation of hepatic progenitors into more differentiated cells (Fig. 1). After 5 days in these culture conditions, the morphology of the cells resembled the cuboidal shapes typical of hepatocytes. In addition, the cells generated expressed markers specific for mature liver cells, including α 1-antitrypsin (AAT), APOA-II, tyrosine aminotransferase, tryptophan 2,3-dioxygenase, factor IX, and the detoxifying enzymes Cyp3A7 and Cyp7A1 as compared with human fetal and

adult hepatocytes (Fig. 5A). Generation of hepatocytes was confirmed by immunostaining analyses showing that differentiating hepatic progenitors expressed nearly homogeneous CK8/18 and that clusters of cells expressed AAT and high levels of albumin (Fig. 5B). FACS analyses revealed that 35% of the cells expressed asialoglycoprotein receptor 1, low-density lipoprotein receptor, c-met, and α 6 integrin (Fig. 5C). Interestingly, the latter two cell surface markers are hallmarks of proliferating hepatic cells *in vivo*.

To further confirm the identity of these hepatocyte-like cells, we transduced undifferentiated hESCs with a recombinant lentivirus carrying the GFP under the control of the human APOA-II regulatory sequences and induced their differentiation. Cells transduced with a lentivector carrying the GFP under the control of the EF1 α promoter were used as a positive control. Flow cytometry analyses revealed that 75% of the control cells were fluorescent 48 hours after transduction, whereas expression of GFP was not detectable when driven by the APOA-II promoter. Cells transduced with the APOA-II-GFP lentivector began to express GFP only after 13 days of differentiation (Fig. 5D,a), confirming the progressive differentiation of endoderm cells into mature liver cells and also demonstrating that hepatic cells generated in our culture system display the physiological regulation of a hepatic-specific promoter. In addition, we tested whether these differentiated cells were functional *in vitro*. Periodic acid-Schiff staining revealed that 60% of the cells could store glycogen (Fig. 5D,b). In addition CK19-positive cells and AFP-positive cells were capable to uptake LDL (Fig. 5D,c, D,d). We also examined uptake and excretion of indocyanine green (ICG), which is a functional characteristic of hepatocytes. ICG-positive cells were visible, and the cells had excluded ICG by 16 hours after its removal from the medium (Fig. 5E,a, E,b). Analysis of the secreted amount of albumin in the culture medium revealed that these cells secreted albumin at a rate of $5.9 \pm 0.7 \mu\text{g}/10^6$ cells/day. Thawed fetal hepatocytes secreted albumin at a rate of $5.6 \mu\text{g}/10^6$ cells/day.

Finally, generated hepatocytes displayed CYP3A7 activity (Fig. 5F).

ESC-Derived Hepatic Cells Are Functional In Vivo. We then investigated the capacity of hepatocytes generated from hESCs to engraft and to differentiate within liver parenchyma. GFP-expressing hESCs were differentiated for 21 days, and the resulting cells were transplanted into the liver uPAxrag2gammac^{-/-} mice. These immunodeficient transgenic mice express the urokinase gene under the control of albumin promoter. This transgene is toxic for hepatocytes, thus it blocks transient liver growth (until the transgene is inactivated in resident

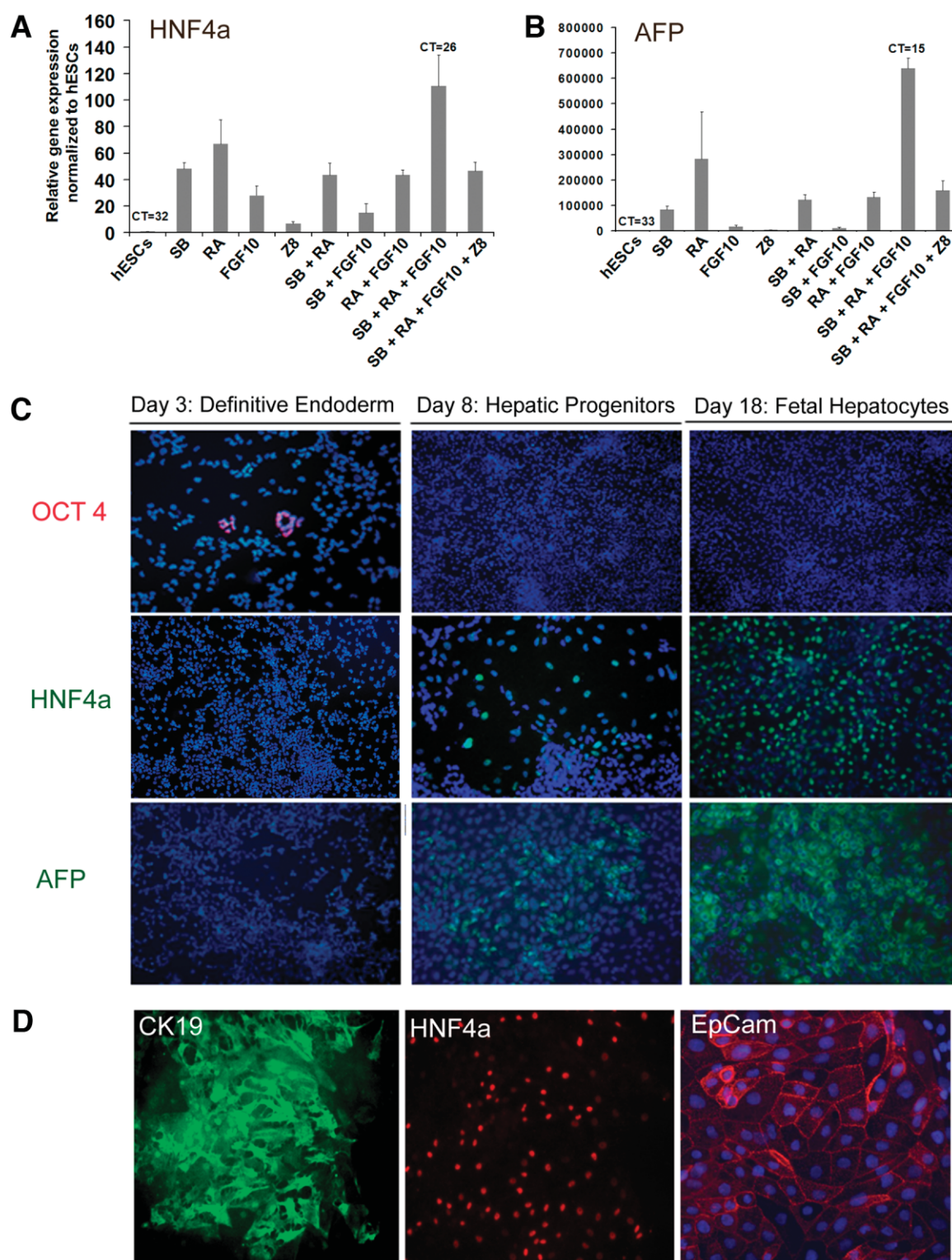


Fig. 4. Definitive endoderm cells generated from hESCs can differentiate into hepatic progenitors. (A, B) DEC were grown for 5 days in CDM/PVA supplemented with a diverse combination of factors (FGF10, RA, activin receptor inhibitor SB54142 [SB], Wnt inhibitor Frizzled-8/Fc chimera [Z8]). Expression of the denoted genes was determined by Q-PCR. CT of PBGD, 23 ± 1 . (C) Expression of pluripotency marker (Oct-4) and hepatic markers (HNF4 α , AFP). Nuclei are highlighted with Hoechst staining (blue fluorescence). (D) Expression of CK19, HNF4 α , and EpCAM in the hepatic progenitors after 8 days of differentiation.

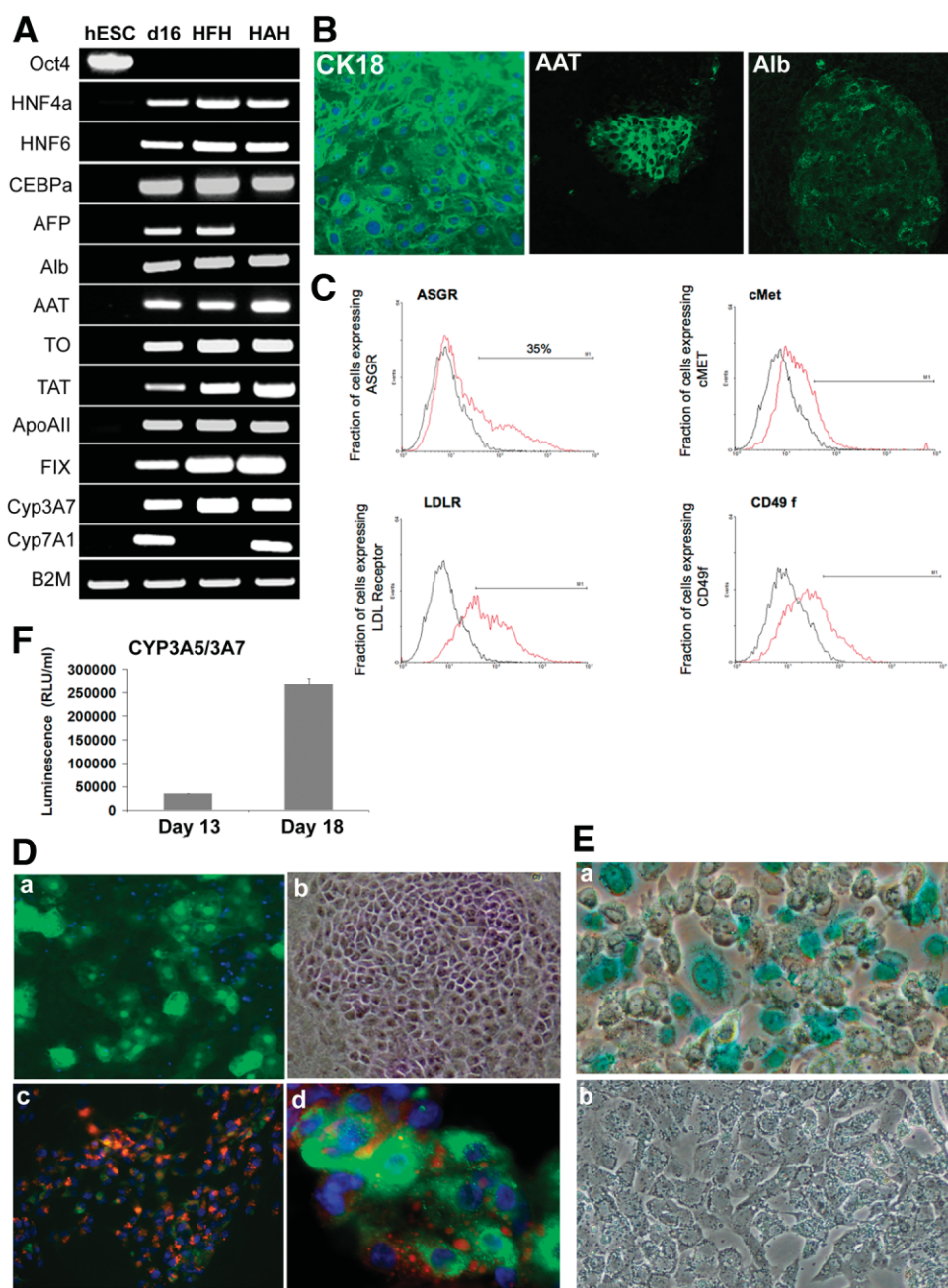


Fig. 5. Hepatic cells generated from hESCs display characteristics and functional properties specific of hepatocytes. (A) RT-PCR analysis of liver cell gene expression in day 18 differentiated cells. Fetal hepatocytes and adult hepatocytes were used as a positive control. (B) Expression of CK18, AAT, and albumin in day 18 differentiated cells. Data are representative of 10 independent experiments. (C) FACS analysis of asialoglycoprotein receptor, HGF receptor (cMET), low-density lipoprotein receptor, and α 6-integrin (CD49f) expression. Data are representative of three independent experiments. (D,a) Expression of APOA-II-GFP after lentiviral transduction of undifferentiated hESCs and 18 days of differentiation. (D,b) PAS stain analysis. Expression of liver cell markers CK19 (D,c) and AFP (D,d) after LDL uptake test is shown. (E,a) ICG uptake in differentiated cells. (E,b) Excretion of ICG from the cells 16 hours later. (F) Hepatocytes generated from hESCs display cytochrome P450 3A7 activity, as assessed by the enzymatic release of free luciferin by cytochrome P450 from an inactive luciferin precursor. The results are expressed as relative light units (RLU) per milliliter of media normalized to the number of cells at day 13 and day 18 of differentiation.

cells), allowing a better engraftment of the transplanted cells. Immunohistochemical analyses revealed the presence of cells expressing human AAT and albumin in the livers of transplanted mice (Fig. 6A-D), confirming that hepatic cells generated from hESCs were capable of en-

grafting in vivo and expressing protein characteristics of hepatocytes. Human cells were distributed throughout the liver, mainly as small and large cell clusters, suggesting that transplanted cells had proliferated and participated in liver growth (Fig. 6A-D). In addition, human AAT and

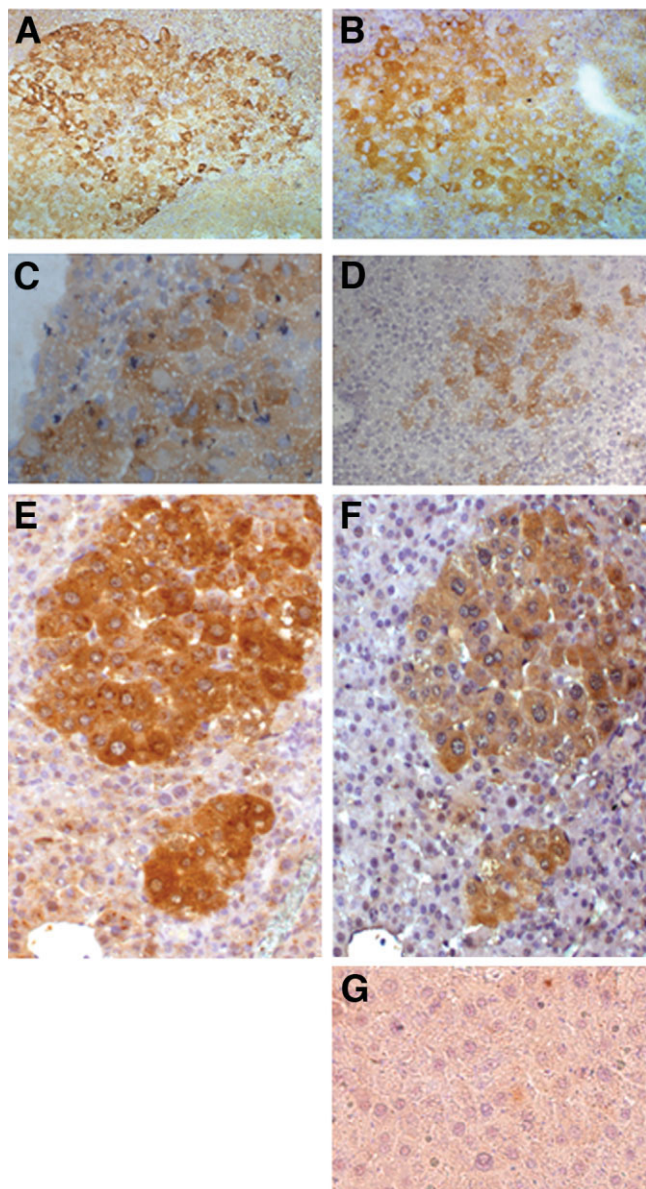


Fig. 6. Engraftment of GFP-expressing hepatic cells into the livers of immunodeficient uPA mice. hESCs were transduced by a GFP-expressing lentivirus, differentiated for 18 days, and then injected into the livers of newborn uPAxrag2gammac^{-/-} mice. The transplanted livers were analyzed for the presence of human hepatocytes 8 weeks after cell infusion by way of immunohistochemical staining. Clusters of (A,B) human AAT-expressing cells and (C,D) human albumin-expressing cells were found in liver sections. The exogenous origin of these clusters was confirmed by costaining for (E) GFP and (F) AAT. (G) Representative field showing no GFP staining in a nontransplanted liver section.

GFP protein were coexpressed in the same cells, confirming the human origin of these cell clusters (Fig. 6E,F). Furthermore, the serum of transplanted animals contained 3 ng/mL human albumin, confirming that the transplanted cells displayed some characteristics of hepatocytes *in vivo*. Finally, histological examination did not reveal the presence of teratomas or intrahepatic tumor,

suggesting that the cell population injected contains only fully differentiated cells.

Generation of Hepatic Progenitors from hiPSCs Using Culture Conditions Developed for hESCs. hiPSCs can be derived from reprogrammed fibroblasts. Therefore, we investigated whether the culture conditions developed to generate hepatocytes from hESCs could also be efficient in differentiating hiPSCs into hepatic cells. Foreskin fibroblasts were reprogrammed in CDM + activin A + FGF2 as described¹⁸ using retrovirus expressing Oct-4, Sox2, KLF4, and cMyc and three of the resulting hiPSCs lines were grown in the culture conditions described above. Immunostaining and Q-PCR analyses revealed that the cells generated under these culture conditions expressed HNF4 α , AFP, and albumin at levels similar to those found in hESCs differentiated using the same culture conditions (Fig. 7A,B). Taken together, these data suggest that our approach developed with hESCs can be used to generate liver cells from hiPSCs.

Discussion

Although several methods are currently available to generate hepatocyte-like cells from hESCs, our approach has two major advantages. It is based on fully defined media devoid of feeder cells and serum, and it also avoids the use of sodium butyrate or dimethyl sulfoxide, both of which are known to affect the epigenetic profile of mammalian cells by respectively inhibiting histone acetylation and increasing DNA methylation.²⁸ Consequently, our method brings clear improvement on existing methods, allowing the differentiation of hESCs into hepatocytes.^{10-13,29}

Furthermore, our protocol follows a process that mimics the progressive differentiation of definitive endoderm cells into hepatic cells during mammalian development. In the first step, hESCs are differentiated into DEC⁺s using a combination of high doses of activin A, FGF2, BMP4, and Ly294002. The use of this PI3K inhibitor to increase endoderm differentiation of hESCs has been shown previously.²¹ However, this study was based on media containing serum, matrigel, and feeders. In addition, inhibition of PI3K in our conditions was not sufficient to block the effect of activin signaling on pluripotency. Consequently, other factors such as BMP4 are required to repress the expression of pluripotent genes and to transform Activin signaling into an inductive signal for endoderm differentiation. These results emphasize the importance to use fully defined culture conditions to define signaling pathways controlling key cell fate choice. In the second step, combination of FGF10 and RA, two factors known to be critical for liver growth and for hepatoblast survival *in vivo*,^{30,31} appear to be essential to drive

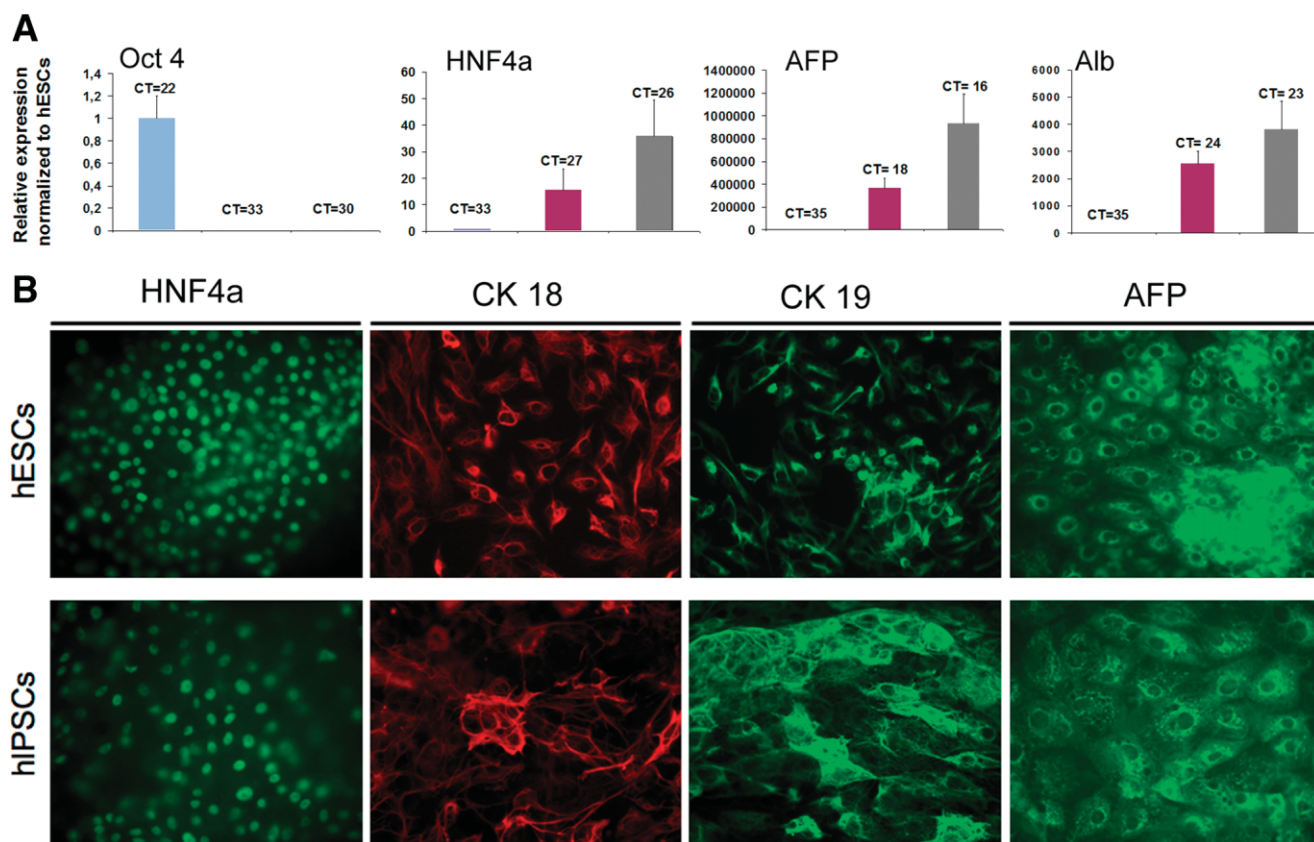


Fig. 7. hiPSCs can differentiate into hepatic cells using culture conditions defined for hESCs. (A) RNAs were extracted after 12 days of differentiation, and expression of the denoted genes was analyzed using Q-PCR. CT of PBGD, 24 ± 1 . (B) Hepatic cells generated from hiPSCs express markers specific for hepatic progenitors (HNF4 α , CK19, CK18, and AFP).

differentiation of endoderm cells into hepatic progenitors coexpressing CK19 and HNF4 α . These two markers are specifically coexpressed by bipotential hepatoblasts at early stages of development, and their expression becomes segregated between biliary epithelial cells and hepatocytes, respectively, during liver organogenesis. However, we also observed that inhibition of activin signaling can increase hepatic differentiation, by contrast to studies in the mouse embryo showing that transforming growth factor β is necessary for liver bud formation.³² However, the function of activin/transforming growth factor β signaling in mouse and human liver development might not be conserved. Alternatively, liver bud specification takes 3–4 days in human versus 12–24 hours in the mouse; therefore, mechanisms happening very quickly during mouse development could become more evident during differentiation of hESCs *in vitro*.

The third step consists in differentiating these hepatic progenitors into hepatocytes while maintaining a proliferative status using a combination of growth factors (FGF4, EGF, and HGF) known to be involved in this process *in vivo*.^{33,34} The cells generated expressed various adult liver-specific proteins, as well as key hepatocyte nu-

clear factors required for controlling the expression of many liver-specific genes. In addition, these cells also exhibited specific function of liver cells. Although these characteristics suggest that hESC-derived hepatocytes are differentiated, our results also suggest that this population is still at the fetal liver developmental stage. Indeed, these hepatocytes retain some immature characteristics, such as expression of AFP, as already reported by others.^{10,14} Further investigations will be required to determine conditions for generating fully mature hepatocytes. Additional inducing factors in combination with high cell-density culture or coculture with other cell types such as endothelial cells might represent a potential solution to overcome this major challenge.

Our study also demonstrates that hESC-derived hepatocytes engrafted efficiently within host liver parenchyma while retaining the ability to proliferate and to display characteristics of normal differentiated hepatocytes. However, low amounts of human albumin in the serum of transplanted mice suggest a lack of correlation between human protein secretion and engraftment efficacy. This has been reported by one of the authors and others in this model.¹⁷ One explanation is that the degree of hepatic cell

differentiation is partial 2 months after transplantation and that engrafted cells produce less than the expected amount of albumin. The xenogenic environment could also underestimate the capacity of human progenitors to fully differentiate.

Tumor formation and abnormal growth remain another major issue when using hESC-derived cells *in vivo*. Indeed, generated populations can be easily contaminated by undifferentiated pluripotent cells, which have the capacity to form teratomas.^{35,36} In addition, the adult environment might not be capable of controlling the proliferative capacity of early progenitors, leading to uncontrolled proliferation. Thus, it was recently reported that hESC-derived AFP-producing cells induced the formation of teratomas.³⁷ Adenocarcinomas were also observed intraperitoneally in analbuminemic rats transplanted with hESC-derived hepatocytes.¹⁴ Importantly, hepatic cells generated using our three-step approach did not produce tumors after transplantation, suggesting that our method induces differentiation of the totality of pluripotent cells.

Although the capacity to generate large quantities of hepatocytes presents obvious advantages for *in vitro* studies such as pharmacotoxicology, the use of terminally differentiated cells might present several drawbacks for cell-based therapy. Indeed, numerous studies in rodents have demonstrated that hepatocytes do not proliferate *in vivo* unless they display a proliferative advantage.³⁸ The use of progenitors/fetal hepatocytes could represent an advantageous alternative that might solve these major limitations. We previously isolated such immature cells during the early developmental stages of human liver and showed that they were able to differentiate and proliferate in rodent livers.¹⁷ Moreover, the proof of principle that progenitors can be used efficiently for cell-based therapy has been clearly established for other organs such as neuronal stem cells in the context of neurodegenerative diseases.³⁹

In any case, mouse models will not allow by themselves to determine whether differentiated cells will be safe for clinical applications. Indeed, the quantity of cells transplanted is limited by the size of the organs, and studies on large animals such as nonhuman primates will be needed to carefully address safety issues.

In conclusion, we have provided the first demonstration that fully defined conditions mimicking liver development results in the generation of functional hepatocytes *in vitro* and *in vivo*. We have also shown that this approach is directly transposable to pluripotent stem cells generated by the reprogramming of adult somatic cells from individual patients. This study opens up the way for the production of liver cells from hESCs or hiPSCs reprogrammed from patients with liver disease. This

should allow the creation of new *in vitro* models to perform drug screening and to develop new therapies for liver diseases.

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